Databases

Database Development Lifecycle (continued)



Summary

1. Case study review
2. Requirements Analysis/Organization
3. Conceptual Modeling
4. Chen Notation
5. Entity-types, Relationship-types, and Attribute-types

Bibliography:

- Connolly, T., Begg, C., Database Systems, A Practical Approach to Design, Implementation, and Management, Addison-Wesley, 4th Edition, 2004. (Chapter 10-12 + 16)
- Sommerville, I., Software Engineering, Pearson Education Limited, 9th Edition, 2011. (Chapter 4)
- Teorey, T., Database Modeling and Design: The Fundamental Principles, 2nd Edition, Morgan Kaufmann, 1994.

Review of the previous class

Definition of the Case Study

CASE STUDY "TechSol" and "Desporto Total Training Center":

"TechSol" is a technology company headquartered in Lisbon, specializing in the development of customized data management and information systems solutions since 2005. Recently, the company's CEO, Teresa Maia, received a proposal from the "Desporto Total Training Center" (CTDT), located in the city of Setúbal, to create a data management system that would optimize the center's operations.

CTDT is a renowned institution in the sports sector, founded in 1990 by former Olympic athlete Carlos Silva. The center's board of directors is composed of President Carlos Silva, Technical Director André Santos, and Training Coordinator Maria Oliveira.

Definition of the Requirements

Desporto Total Training Center (CTDT)

Database System Development Process

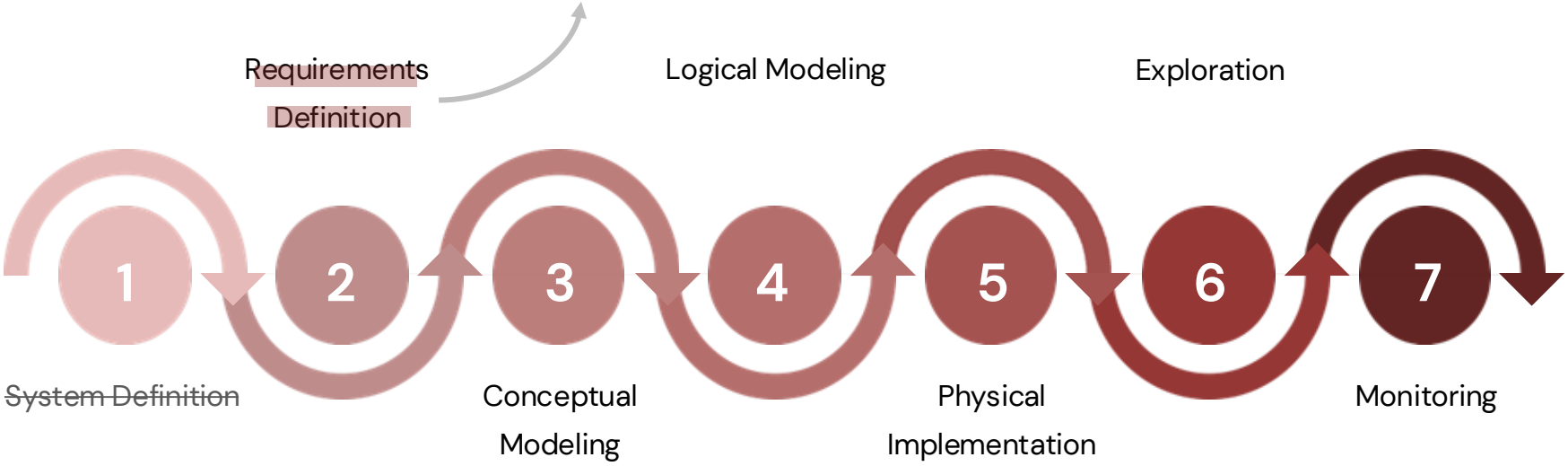
Requirements Gathering

General Requirements Document

No.	Date and Time	Description
1	22/02/2023 17:00	All athletes of the center must be registered with a unique identification number (6 digits).
2	22/02/2023 17:00	Athletes' personal data must be stored, such as full name, date of birth, address, phone number, email, tax identification number, etc.
3	22/02/2023 17:00	In terms of profession type, only athletes, healthcare professionals, and coaches are to be considered.
4	22/02/2023 17:00	Each specialty must be characterized by an identification code and a detailed description.
5	23/02/2023 17:00	At any time, it must be possible to obtain the list of healthcare professionals for a given specialty.
6	22/02/2023 17:00	Only one consultation may take place at a time in a given consulting room and time slot, avoiding overlaps.
7	23/02/2023 17:00	During a consultation, professionals may schedule exams.
8	24/02/2023 17:00	At any time, it must be possible to obtain the consultation history of a given athlete.

Database Development Lifecycle

Survey, analysis, and organization of requirements regarding the part of the organization that will be supported by the database system (DBS). The requirements must be able to describe which data is intended to be stored and the possible relationships between them.



Database Development Lifecycle:
Requirements definition

In a requirements gathering process, we must take into account some important tasks, namely:

– Identify the essential information of the real problem.

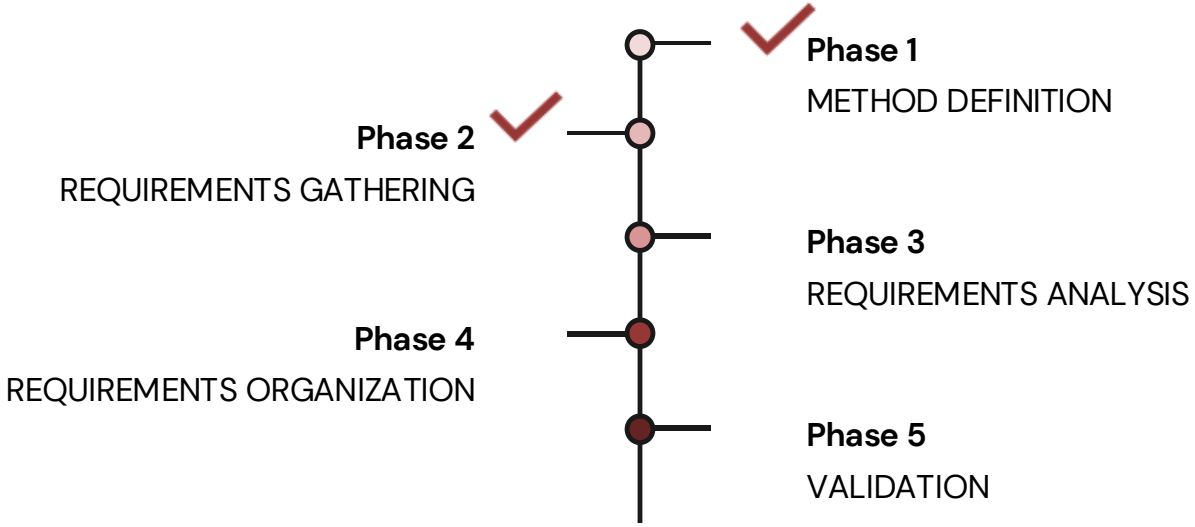
– Remove redundant or irrelevant information for the problem at hand.

– Clarify collected information that is unclear.

– Fill in any gaps or missing information found.

– Be able to identify the data and the processes that work with them.

Database Development Lifecycle:
Requirements definition

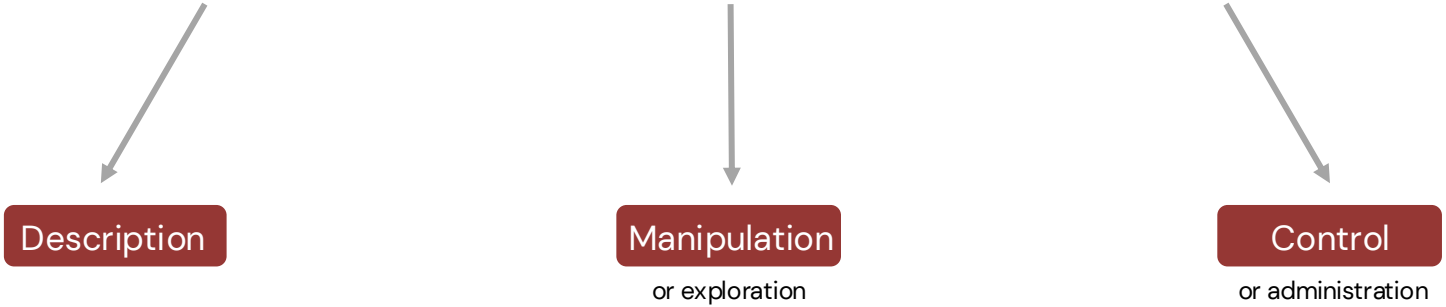


PHASE 2: Requirements Definition

Analysis and Organization of Requirements

After carrying out the survey, it is important to perform a critical analysis of the requirements, one by one, to ensure that there are no errors, inconsistencies, redundancy, or ambiguities.

During the analysis, analysts can begin organizing the requirements according to these three areas of work:



PHASE 2: Requirements Definition

Analysis and Organization of Requirements

Description

Requirements associated with the creation of objects in the DB

DDL (Data Definition Language)

CREATE

ALTER

DROP

TRUNCATE

COMMENT

RENAME

Manipulation

Requirements associated with the interaction with existing data in the DB

DML (Data Manipulation Language)

SELECT

INSERT

UPDATE

DELETE

MERGE

CALL

LOCK TABLE

Control

Requirements associated with the control of different user profiles (access/operation permissions)

DCL (Data Control Language)

GRANT

REVOKE

PHASE 2: Requirements Definition

Analysis and Organization of Requirements

EXAMPLES:

DESCRIPTION REQUIREMENTS:
DR1 - All athletes at the center must be registered with a unique identification number (6 digits).
DR2 - Personal data of the athletes must be stored, such as full name, date of birth, address, phone number, email, tax identification number, etc...
(...)

MANIPULATION REQUIREMENTS:
MR1 – At the end of the year, it must be possible to know the number of training sessions conducted by each coach.
MR2 – It must be possible to access an athlete's procedure history.
(...)

CONTROL REQUIREMENTS:
CR1 – Only athletes can add feedback about the training sessions.
CR2 – Only doctors are allowed to schedule exams.
(...)

PHASE 2: Requirements Definition

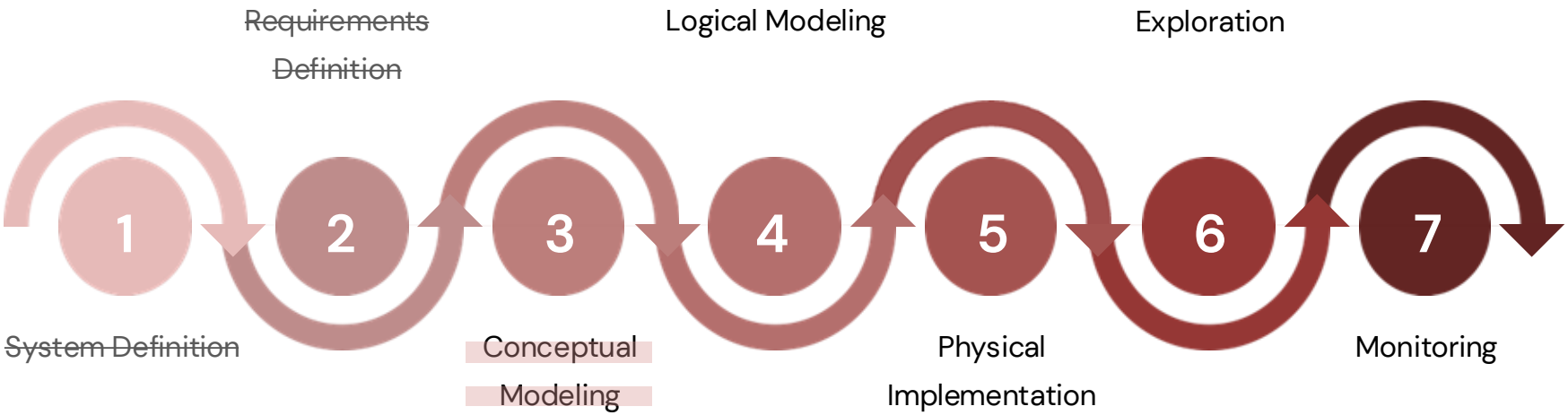
Validation

The validation of requirements ensures that the requirements are correct, complete, and reflect the users' needs.

Why validate?
- Avoid errors in subsequent phases;
- Reduce “rework”;
- Ensure alignment with the business;

Challenges
The definition of requirements is a demanding and resource-consuming step, which requires domain knowledge, patience, and frequent reviews to correct errors and complete information.

Database Development Lifecycle



Database Development Lifecycle

Connolly and Begg Methodology (2005)

Connolly and Begg proposed a methodology to support the analysis and modeling of database systems that integrates three stages of the database lifecycle:

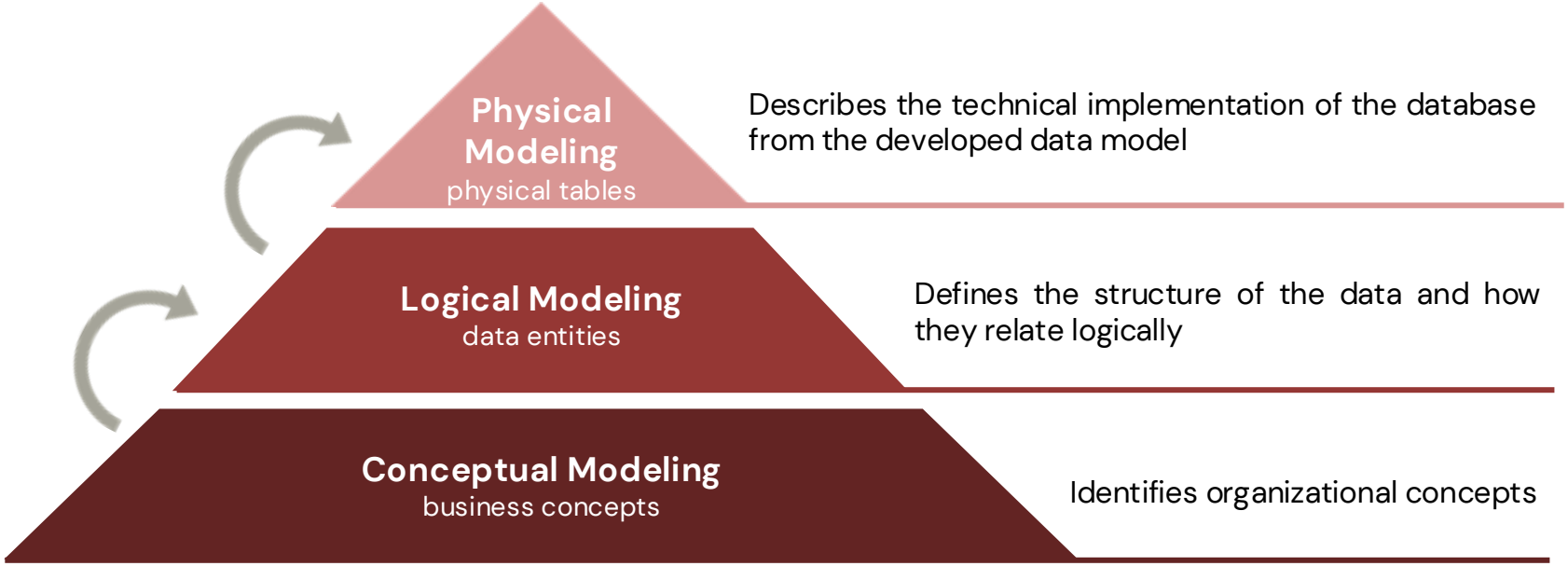
Part I — Conceptual Database Design
Process of constructing a model of the information used in an organization, independent of any physical considerations.

Part II — Logical Database Design
Process of constructing an information model based on a specific data model, but independent of any database management system (DBMS) or physical considerations.

Part III — Physical Database Design
Process of defining the implementation of the database in secondary storage systems, including the description of storage structures and access methods used to ensure efficient access to data.

Database Development Lifecycle

The development project of a database has three main modeling phases:



Database Development Lifecycle: Conceptual Modeling

The conceptual design of a database consists of constructing an information model used in an organization, independent of all physical considerations, such as:

- Target database management system (DBMS)
- Application programs
- Programming languages
- Other implementation aspects

Database Development Lifecycle:
Conceptual Modeling

One View, One Model

A user view (or usage view) is the set of data necessary for a particular user to make decisions or perform tasks, typically corresponding to a functional area of the organization.

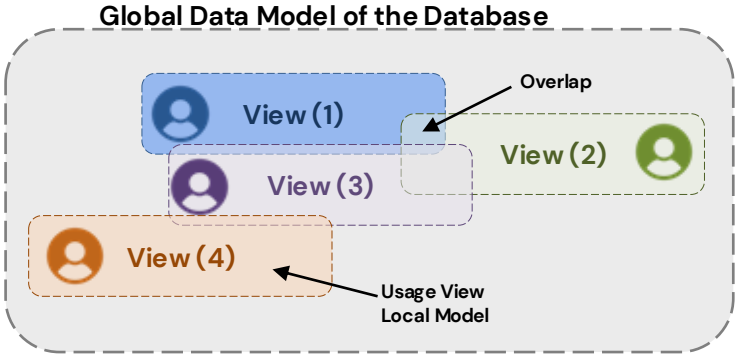
User views can be identified during the requirements gathering phase through methods such as:
– Analysis of data flow diagrams
– User interviews
– Analysis of procedures, reports, and forms
– Observation of the organization's operations

• In the conceptual data model, each user view to be modeled is designated as a local conceptual data model.

Database Development Lifecycle:
Conceptual Modeling

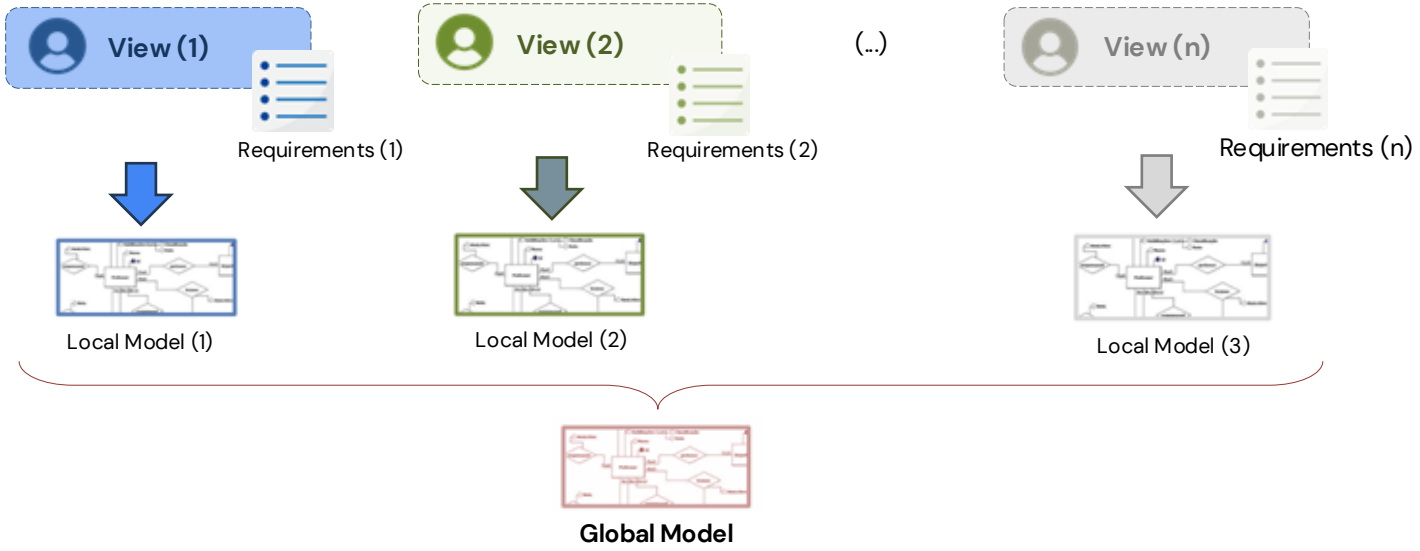
Local Data Models

- Represent the data from a user's view
- Correspond to a specific part of the problem and the business model
- Facilitate the modeling of a user's activities
- Constitute a partial view of the global schema



Database Development Lifecycle:
Conceptual Modeling

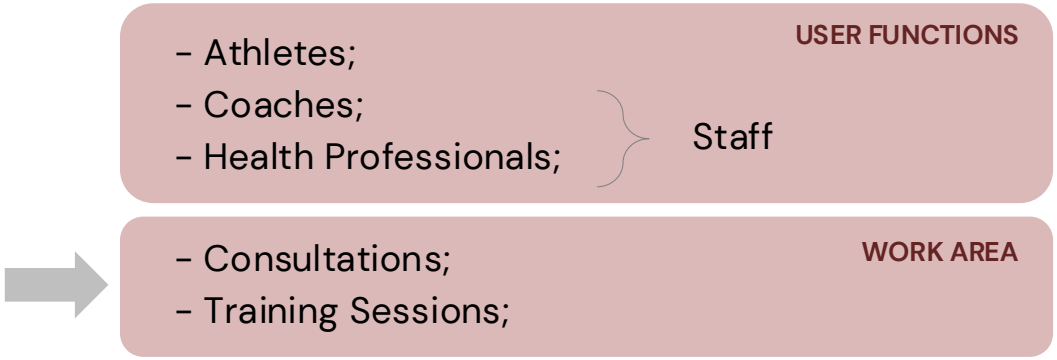
View Integration



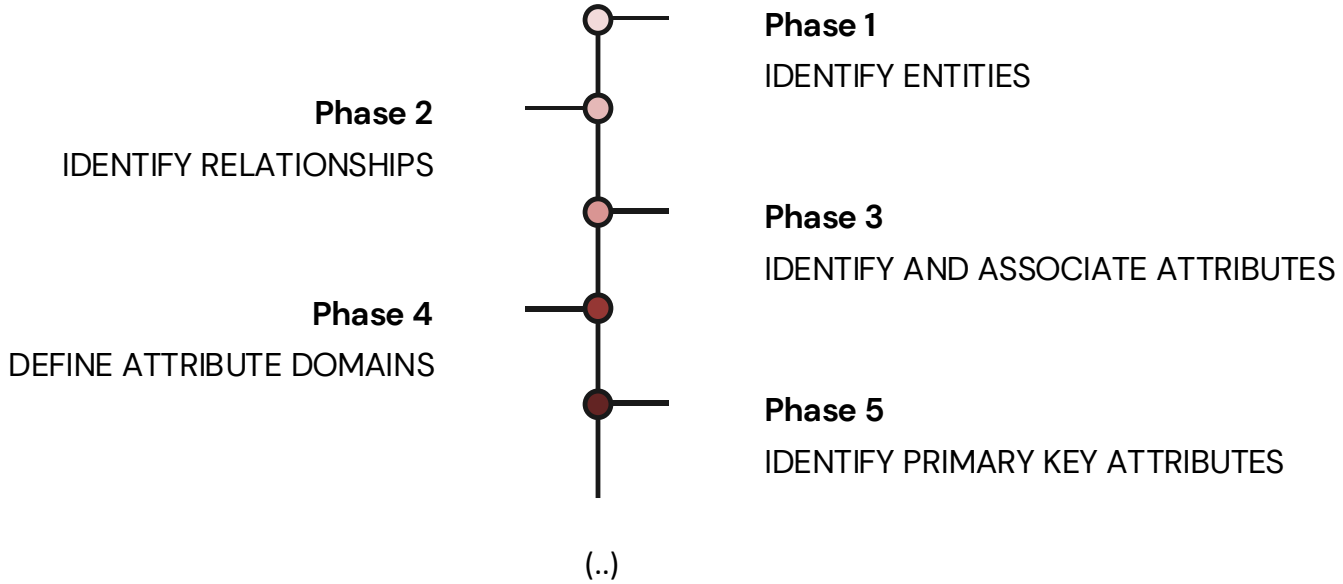
Usage Views

Example

According to the requirements defined for the training center case study, the following usage views were identified:

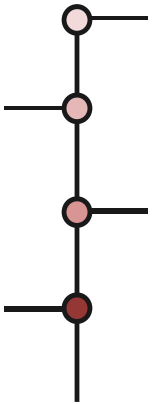


Database Development Lifecycle:
Conceptual Modeling



Database Development Lifecycle:
Conceptual Modeling

(..)



Phase 6
CONSIDER THE USE OF ENHANCED MODELING CONCEPTS (optional)

Phase 7
CHECK FOR REDUNDANCY

Phase 8
VALIDATE USER TRANSACTIONS

Phase 9
REVIEW AND VALIDATE THE CONCEPTUAL MODEL

PHASE 3: Conceptual Modeling

- There are different notations and methodologies in data modeling.
- A high-level data model normally used in conceptual/logical DB design, and the one we will use throughout the classes, is based on the concepts of the **Entity-Relationship (ER)** model.
- A conceptual data model is supported by documentation, such as ER diagrams and a data dictionary, produced throughout the development of the model.
- Currently, there is no standard notation for an ER model, but most books dedicated to this topic tend to use one of two conventional notations:

Chen Notation

Crow's Feet Notation

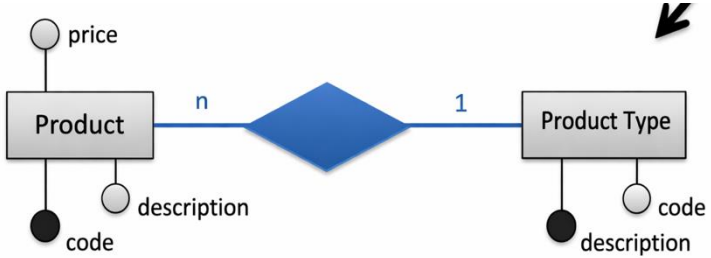
PHASE 3: Conceptual Modeling

The Entity-Relationship Approach

Central Concepts:

– Entity;
– Relationship;
– Attribute.

ER Diagram



Each of these concepts will be analyzed in detail next, according to the steps previously presented.

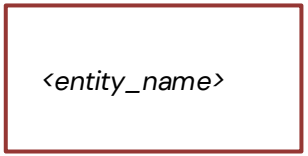
PHASE 3: Conceptual Modeling

Identify entities

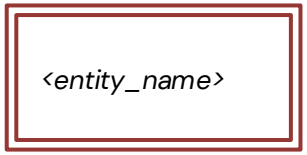
Objects or concepts from the real world with the same properties that have an "independent existence" (physical or conceptual).

How to identify entities?
- Examine the description requirements;
- Nouns, names;
- Objects such as people, places, interests, etc.;

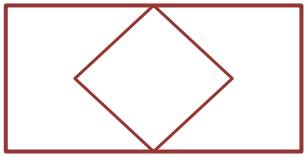
The analysis is subjective, that is, different approaches can generate different entities for the same problem.



Entity



Weak Entity



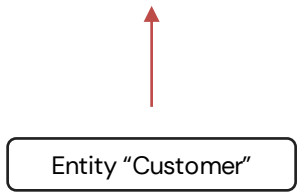
Associative Entity

PHASE 3: Conceptual Modeling

Identify entities

Let us consider, for example, the following description requirement, previously raised by the analysts:

RD1: "A **customer** record must contain a unique identifier."



PHASE 3: Conceptual Modeling

Identify entities

The entities of a conceptual model must be well documented so that their role in the database is understood. They must have clear and meaningful names, and their descriptions must be recorded in the data dictionary, including, whenever possible, the number of occurrences and any synonyms.

Entities Table (example for the case study):

Entity	Description	Occurrence
Training	Planned exercise session to improve skills or physical conditioning.	The training sessions include an athlete and a coach.
Coach	Person responsible for guiding and supervising athletes during training.	Each coach has an associated modality.
Equipment	Tools, devices, or instruments used during training to assist in physical exercises.	Coaches can request equipment during training sessions.
…	…	…

PHASE 3: Conceptual Modeling

Identifying relationships

Representation of associations/connections between two or more entities. Each relationship is described by a name that represents its function (whenever possible this name should be unique in the model).

How to identify relationships?
- Examine the description requirements;
- Verbs or verbal expressions;

Most relationships are binary, involving two entities, but there may be more complex relationships with three or more entities or recursive relationships involving the same entity.



Relationship



Relationship associated with a weak entity

PHASE 3: Conceptual Modeling

Identifying relationships

Let us consider, for example, the following description requirement, previously raised by the analysts:

*RD2: All **sales** records must include a reference to the **customer** who made it.*



PHASE 3: Conceptual Modeling

Identify relationships

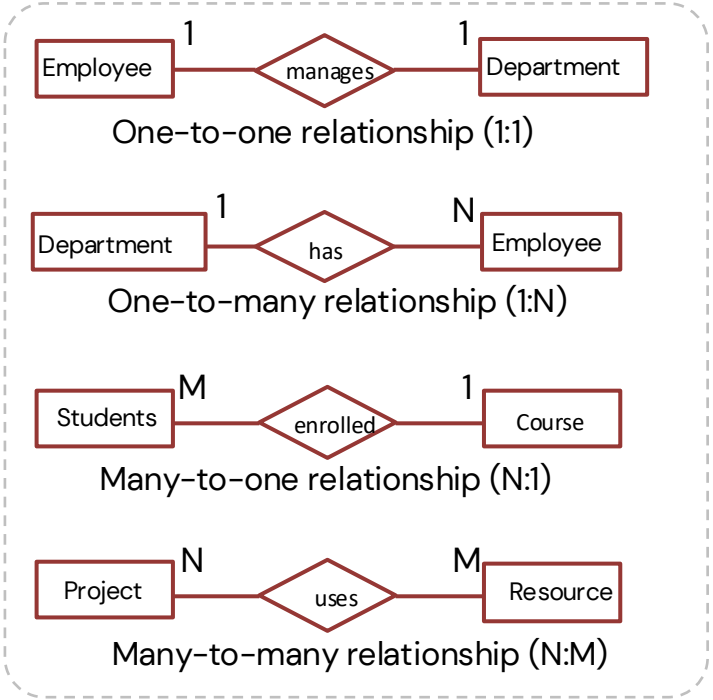
Cardinality

Cardinality indicates the number of occurrences of one entity that can be associated with another through a relationship.

- **1–1 (One to One)**
  Each occurrence of one entity is associated with only one of the other

- **1–N (One to Many)**
  One occurrence can be associated with several of the other entity

- **N–N (Many to Many)**
  Several occurrences can be associated with each other



PHASE 3: Conceptual Modeling

Identify relationships

Participation

optional/partial
Not all occurrences of the entity need to participate

mandatory/total
All occurrences of the entity must participate in the relationship

Example:



One-to-many relationship with optional participation of the Client entity and mandatory participation of the Purchases entity

A Client may have no Purchases but a Purchase must always have an associated Client

Min/max notation

Defines the minimum and maximum number of participations of an entity in a relationship.
0 (zero): optional association;
1 (one): mandatory association.



A Client may have made at least 0 and at most N purchases but a Purchase has at least 1 and at most 1 client (1 and 1 only).

PHASE 3: Conceptual Modeling

Identify relationships

The documentation about a relationship should include the following elements:

– Name of the entities involved.
– Designation of the relationship.
– Cardinality –1:1,1:N,N:M.
– Participation – P:P, P:T, T:T.

Relationship Table (example for the case study):

Entity A	Entity B	Relationship	Cardinality	Participation
Coach	Training	leads	1:N	T:T
Training	Equipment	requires	N:N	P:P
Plan	has	exercises	N:N	T:P
...	...	...	...	...

PHASE 3: Conceptual Modeling

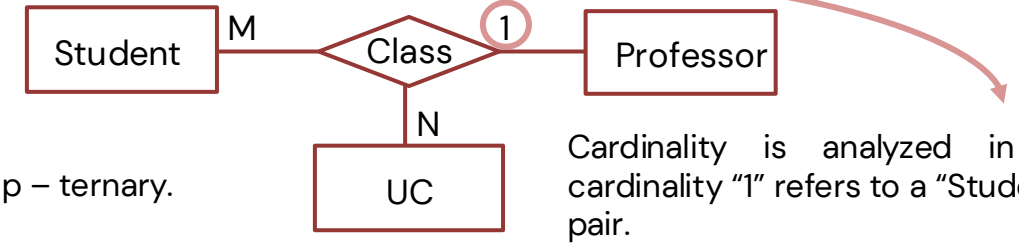
Identifying relationships

The number of participants in a type of relationship is called the degree of that relationship. Therefore, the degree of a relationship indicates the number of entity types involved in that relationship.

A relationship of degree two is called binary. This is the most common degree of relationship.
Example: An employee drives a vehicle.

Relationships with a degree higher than two are called complex relationships (ternary, quaternary, etc.)

Example:



Three-degree relationship – ternary.

Cardinality is analyzed in pairs. The cardinality "1" refers to a "Student" and "UC" pair.

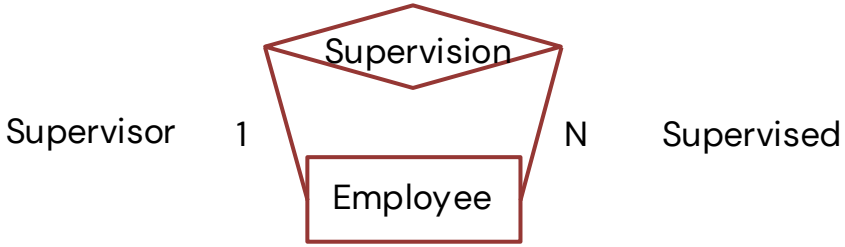
PHASE 3: Conceptual Modeling

Identify relationships

Recursive Relationships

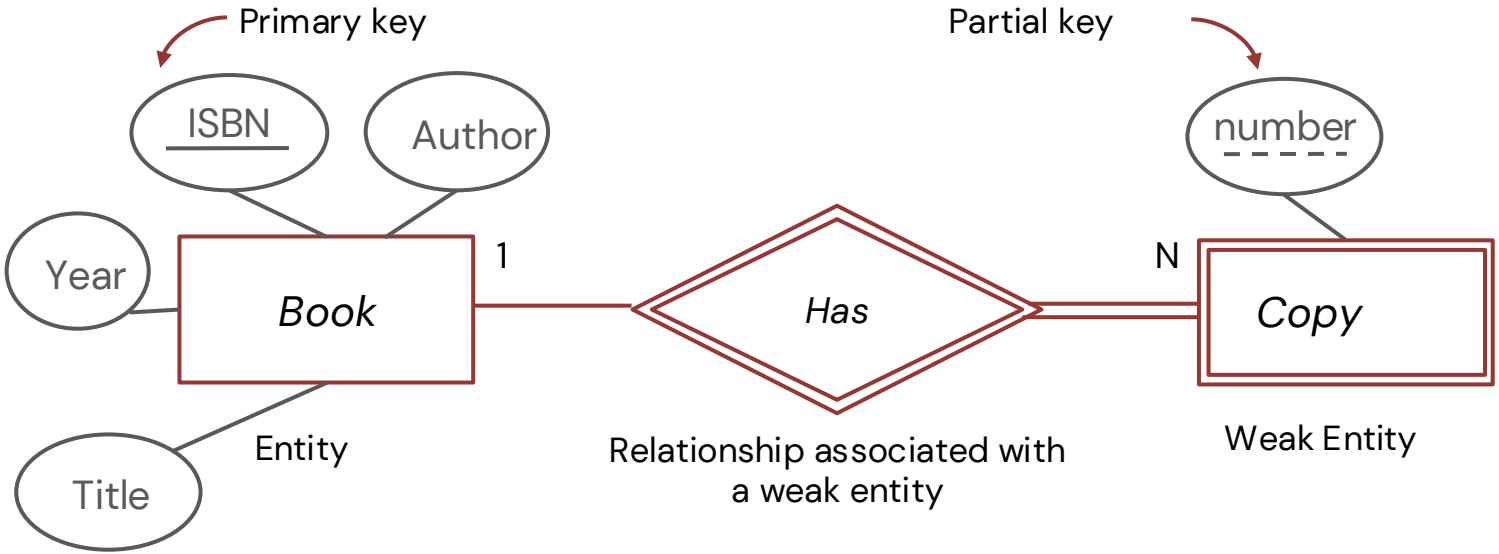
A type of relationship in which the same entity type participates more than once with different roles.

Example: Consider the example of a recursive relationship called “Supervision”, which represents an association of a Supervisor where the Supervisor is also an employee. The entity type “Employee” participates twice in the “Supervision” relationship; the first participation as Supervisor and the second participation as the employee who is Supervised.



PHASE 3: Conceptual Modeling

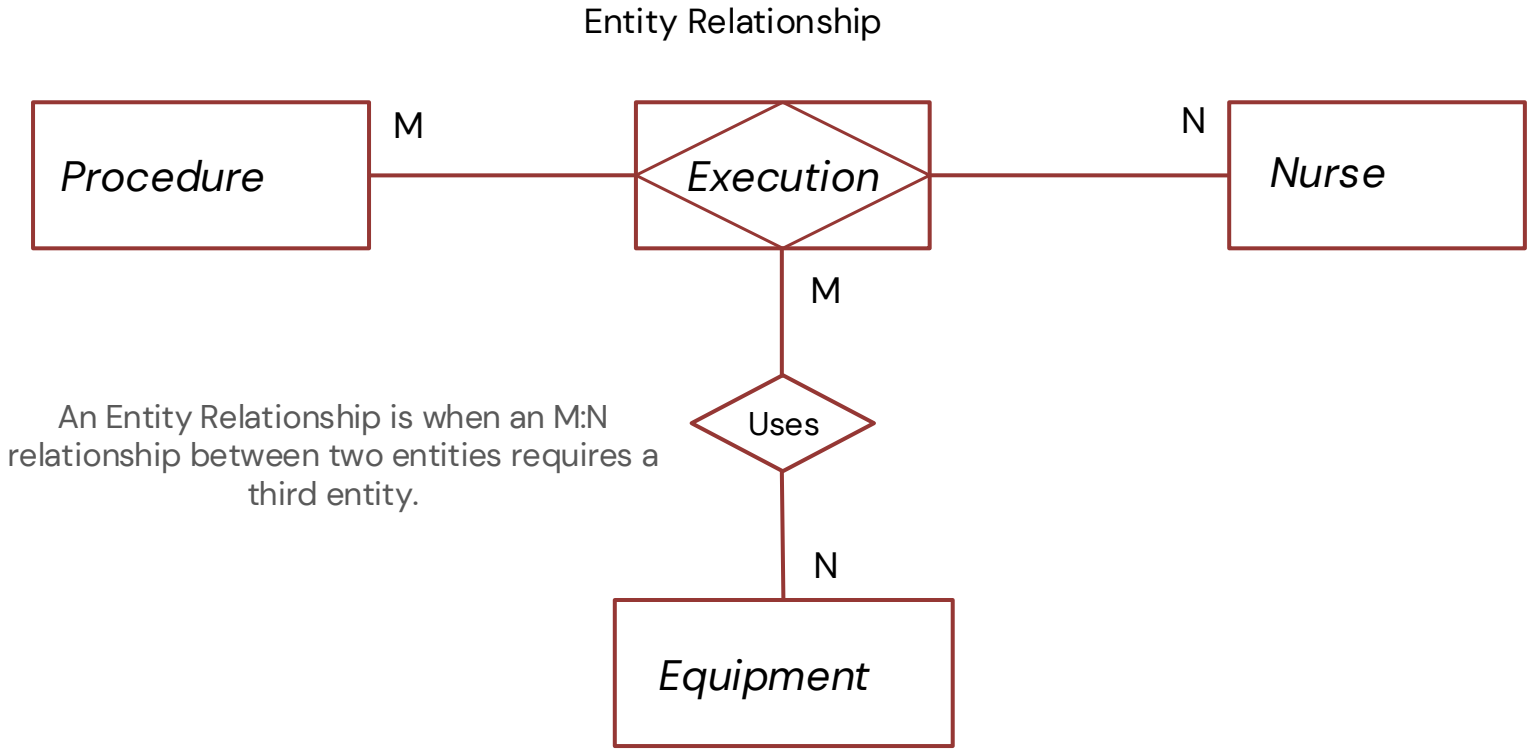
Entities and Weak Relationships



A Weak Entity is always related through a weak relationship and has total/mandatory participation in that relationship,

PHASE 3: Conceptual Modeling

Entity Relationship



An Entity Relationship is when an M:N relationship between two entities requires a third entity.

PHASE 3: Conceptual Modeling

Identify and associate attributes

This step consists of identifying the facts (attributes) that describe the entities and relationships in the model.

An attribute is a property of an entity or relationship. Attributes contain values that describe each occurrence of an entity and represent the main part of the data stored in the database.

How to identify attributes
- Analysis of description requirements.
- Identification of names, properties, and characteristics
- Observation of entities and relationships

Best practices
- Assign clear and meaningful names
- Ensure that the meaning and content are understandable

PHASE 3: Conceptual Modeling

Identify and associate attributes

Let us look, for example, at the following description requirement, previously gathered by the analysts:

Entity "Client"

RD3 – The record of a **client** must contain a unique identifier (number), the full name, a postal address (Street, City, and Postal Code), profession, date of birth, taxpayer number, a list of phone numbers, email, and the total amount of purchases made in the store up to the present moment.

Attributes (in green)

PHASE 3: Conceptual Modeling

Identify and associate attributes

Types of attributes

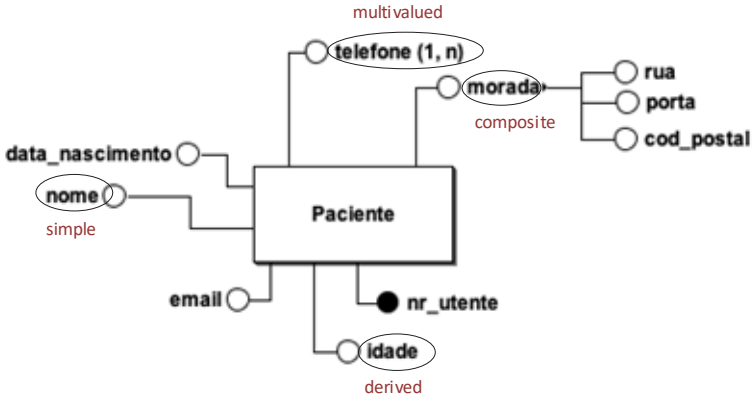
In the process of identifying attributes, we can find different types:

Simple Attributes
• Basic characterization elements
• Cannot be divided

Composite Attributes
• Formed by several simple attributes
Example: Address (Street, Postal Code, City)

Derived (Calculated) Attributes
• Obtained/calculated from other attributes
• Not always presented in data models,
Example: Age (from date of birth)

Multivalued Attributes
• Their content consists of more than one value.
Example: Customer phone numbers



PHASE 3: Conceptual Modeling

Identify and associate attributes

To characterize an attribute, we need to record the following information: the name and characterization of the attribute, data type and length (domain), whether it can have null values or not, in the case of a composite attribute, which simple attributes make it up, when the attribute is derived, indicate how it is calculated, and indicate when an attribute is multivalued.

Attribute Table (example for the case study):

Entity	Attribute	Description	Data Type and Size	Null (Y/N)	Multivalued (Y/N)	Primary Key	...
Coach	Professional_Number	Coach identifier	6 var characters	N	N	Yes	
	Name	Name	100 var characters	N	N	No	
	....	....	....	....	....	....	
....	....	....	....	....	....	....	

PHASE 3: Conceptual Modeling

Identify and associate attributes

Attribute Domains

The domain of an attribute is the set of possible values that an attribute can assume.

What it should include
- Allowed values
- Data type
- Size and format

Examples
- Age → INT (0–120)
- Name → VARCHAR(100)
- Date of Birth → DATE

PHASE 3: Conceptual Modeling

Identify and associate attributes

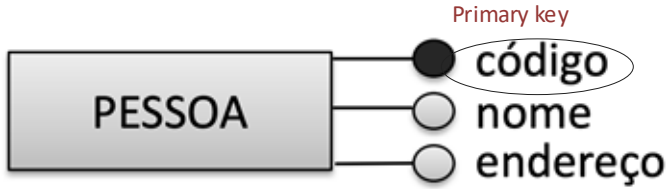
Key Attributes

This step consists of identifying the candidate keys of an entity and selecting the primary key. Each entity must have an identifier;

Candidate Key
- Attribute or minimal set of attributes that uniquely identifies an instance of the entity
- There may be several

Primary Key
- Candidate key chosen to identify the entity
- Must be unique and not null

Alternate Keys
- Remaining candidate keys not chosen as primary
- Also referred to as secondary keys



PHASE 3: Conceptual Modeling

Identify and associate attributes

Key Attributes

In the process of choosing a primary key from a set of candidate keys, we can follow some guidelines, that is, choose the candidate key:

– with the smallest set of attributes.

– with fewer possibilities of having its values modified.

– with less probability of losing its uniqueness in the future.

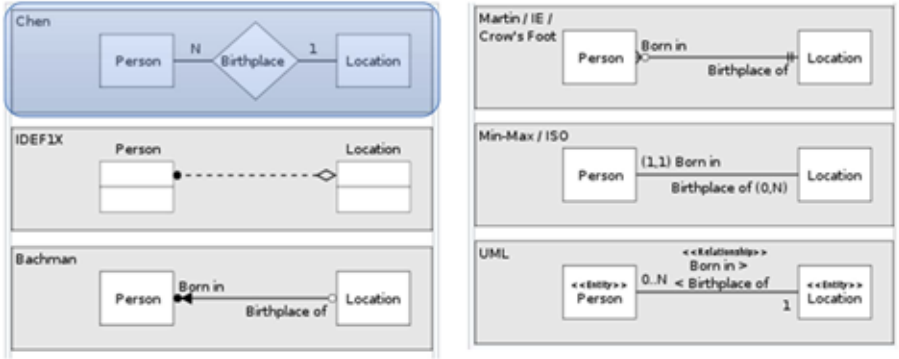
– of smaller size.

– more "convenient", that is easier to use.

PHASE 3: Conceptual Modeling

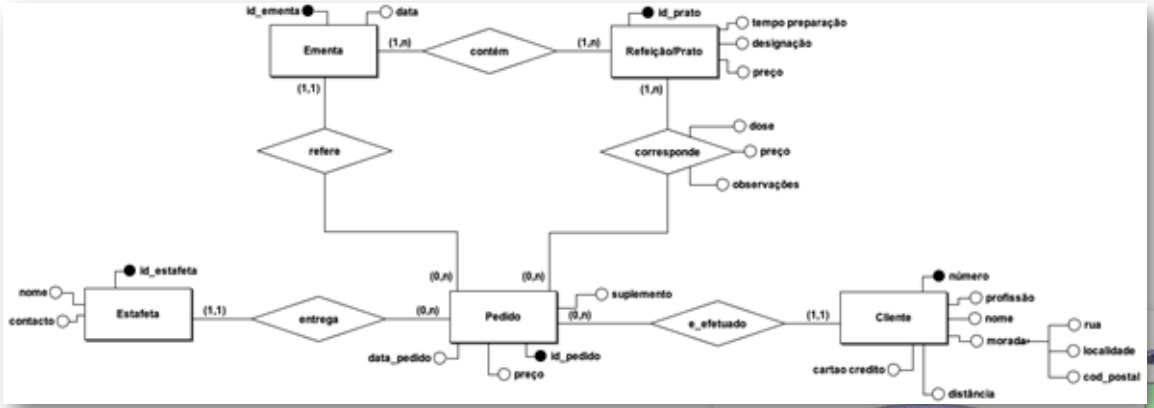
Draw the ER Diagram

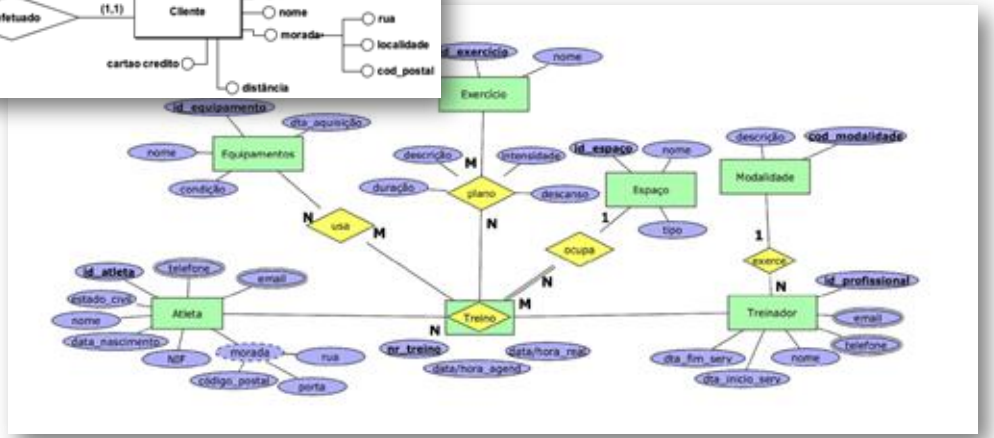
The purpose of drawing an ER diagram is essentially to represent, from a conceptual point of view, the user's view of the organization.
As already mentioned, the construction of an ER diagram can be supported by various notations. Most notations are very similar. Although they may vary in terms of graphic elements, their representation semantics are very similar.



PHASE 3: Conceptual Modeling

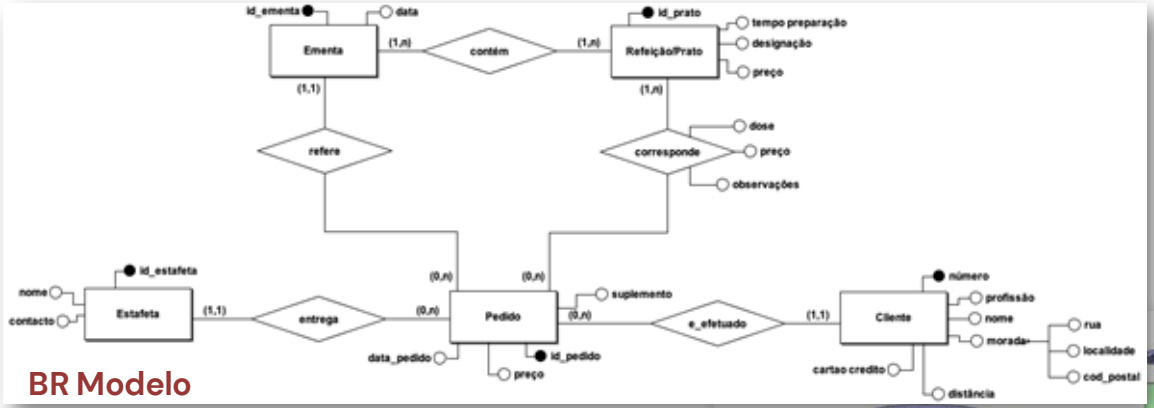
Examples of ER Diagrams





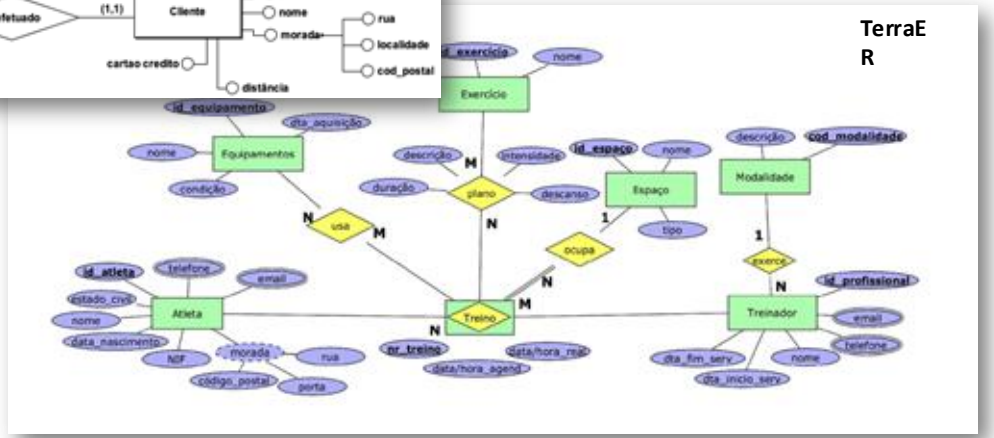
PHASE 3: Conceptual Modeling

Modeling Tools



BR Modelo

http://sis4.com/brModelo/brModelo/



PHASE 3: Conceptual Modeling

ER Diagram

Case Study

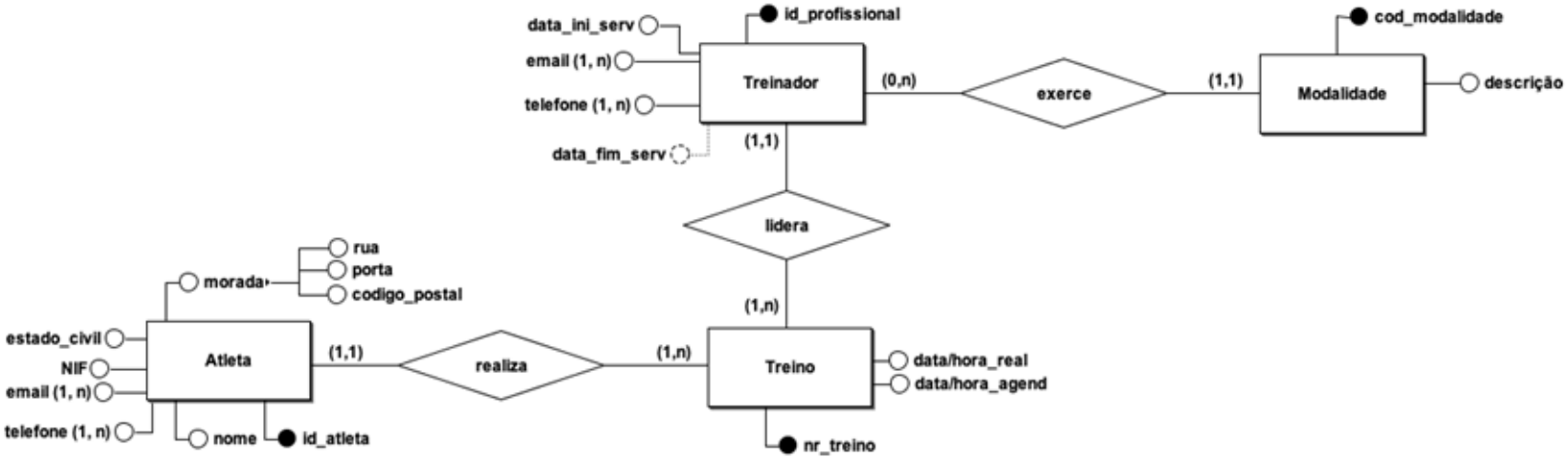
Question 6: Construct an E-R Diagram, using Chen Notation, capable of accommodating the following requirements:

- Each athlete at the center must be registered with a unique 6-digit athlete ID.
- It is necessary to store demographic data of the athlete, namely name, date of birth, address, phone number, email, taxpayer number (NIF), ID, marital status, etc.
- An athlete can be trained by several coaches.
- A coach must be registered with their professional number, name, service start date, end date, phone contact, and email address.
- A coach practices one and only one sport.
- Each sport at the center must be characterized by the sport code and its description.
- The training record must include the training number (unique identifier), the date and time of scheduling, start date and end date of the training, the associated sport, etc.

PHASE 3: Conceptual Modeling

ER Diagram

Training View A



PHASE 3: Conceptual Modeling

Model Review

- This is the last step of the first modeling phase and consists of reviewing the data model developed with the user in order to ensure that the model is a true representation of the user's view of the organization.
- The conceptual data model includes ER diagrams and the respective supporting documentation, including its description – the data dictionary.
- If any anomaly is detected in the model, it is necessary to take the necessary measures for its correct adaptation to the intended usage perspective. It may be necessary to repeat some of the steps previously carried out.

University of Minho

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Databases



The Life Cycle of Database Development (continued)

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